Date	15 august2026
Team ID	…..
Project Title	**India's Agricultural Crop Production**
Maximum Marks	3 Marks

Project Proposal (Proposed Solution)

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	**Visualize Trends:** Analyze historical data to identify long-term crop production patterns, seasonal fluctuations, and regional productivity across Indian states. **Empower Stakeholders:** Guide farmers on crop choice, analysts on policy, and investors on high-growth areas. **Build Dashboards:** Create interactive Tableau visual reports and analytical stories. **Optimize Yields:** Identify high-performing crops and low-yield regions to lower risks and focus resources.
Scope	**Data Preparation:** Clean and standardize multi-year Indian crop data from CSV/Excel/SQL sources. **Coverage:** Analyze major Indian states, union territories, and primary cropping seasons (Kharif, Rabi, etc.). **Visual Development:** Design dynamic Tableau maps, filters, and interactive visual dashboards.

	Testing & Documentation: Optimize dashboard performance and document key insights for presentation.
Problem Statement	
Description	Analyzes two decades of Indian agricultural data (1997–2021) to map regional productivity and crop performance. Cleans and transforms complex multi-year agricultural datasets into standardized analytical formats. Leverages interactive Tableau dashboards to visually track seasonal yield fluctuations and regional disparities. Evaluates trends across diverse cropping seasons (Kharif, Rabi, Whole Year) and major Indian states.
Impact	For **Farmers:** Minimizes financial risk by providing data-backed recommendations for high-yielding seasonal crops. For **Policy Analysts:** Enables targeted government resource allocation and subsidies by highlighting low-yield districts. For **Agribusinesses:** Reduces market uncertainty by revealing long-term crop growth patterns and investment opportunities. For **Data Accessibility:** Converts complex tabular data into intuitive visual stories for faster decision-making.
Proposed Solution	

Approach	• **Data Ingestion & Cleaning:** Aggregate multi-decade raw datasets (1997–2021), clean duplicate/null records, and normalize key variables such as production quantity, cultivated area, and yield metrics. • **Exploratory Analytics:** Perform statistical analysis to categorize regional performance, identify dominant crop varieties, and isolate long-term trends across different agricultural zones. • **Visual Data Architecture:** Build custom dynamic calculations and hierarchical data structures within Tableau to support cross-filtering between states, districts, seasons, and years. • **Dashboard Design & Deployment:** Construct specialized interactive stories and drill-down visual interfaces tailored to farmers, policy analysts, and agribusiness investors.
Key Features	• **Geospatial Yield Mapping:** Interactive state and district maps displaying crop density, yield distribution, and regional production highlights. • **Multi-Year Trend Analysis:** Line and area charts tracking production volume and land usage over 24+ years to map growth trajectories. • **Season-Wise Drill Downs:** Comparative visual matrices isolating Kharif, Rabi, Summer, and Whole Year cropping patterns. • **Interactive Filtering:** Cross-functional KPI cards, drop-down filters, and top N state/crop visual summaries for dynamic stakeholder exploration.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	2 x NVIDIA V100 GPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy
Development Environment	IDE, version control	Jupyter Notebook, Git
Data		
Data	Source, size, format	Kaggle dataset, 10,000 images